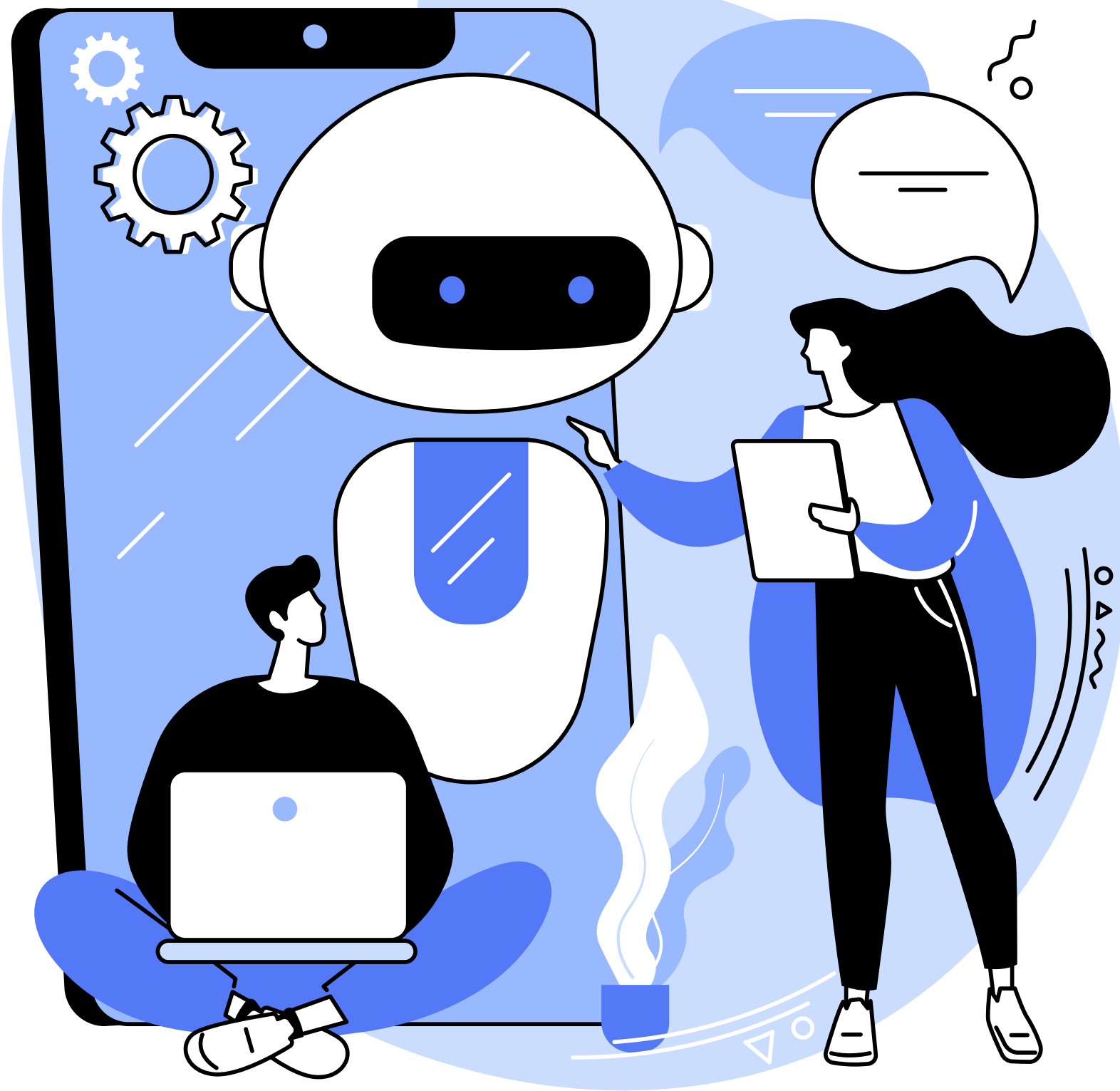
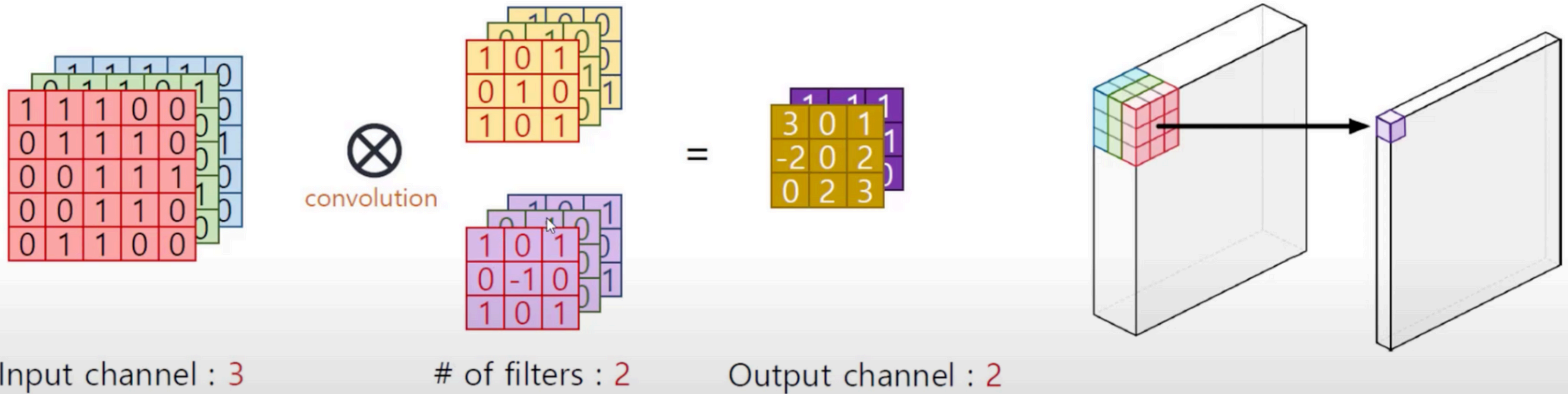
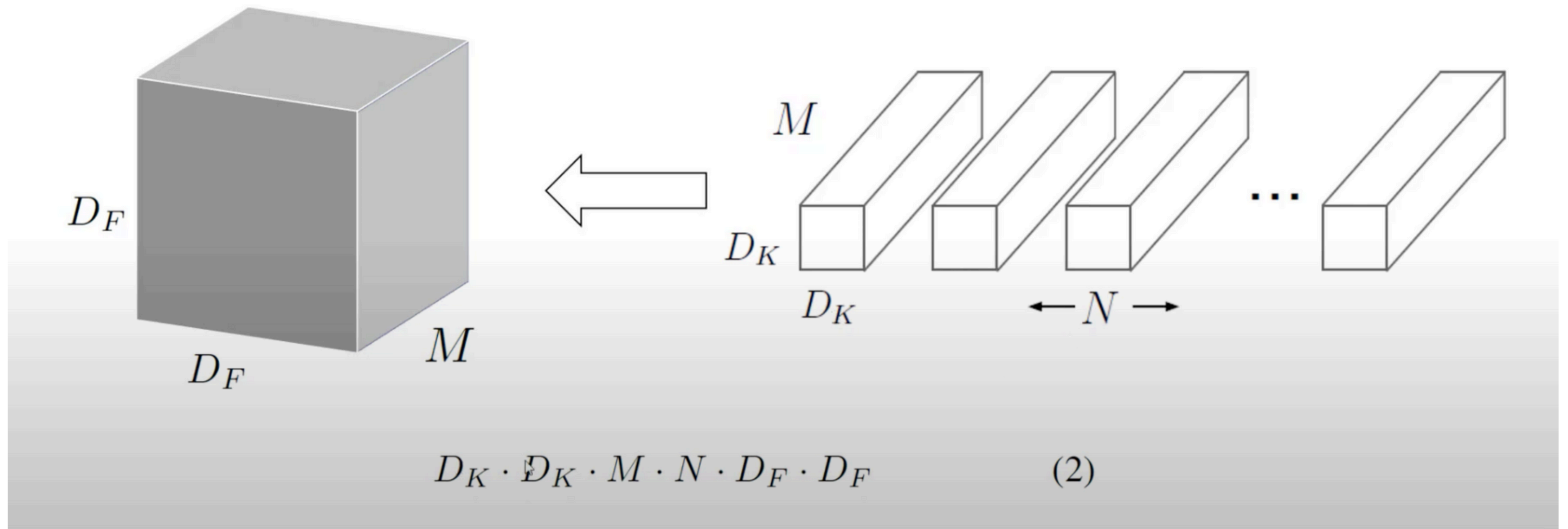




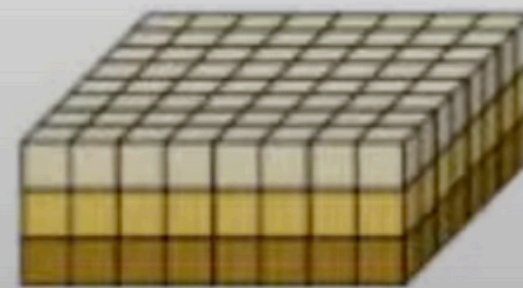
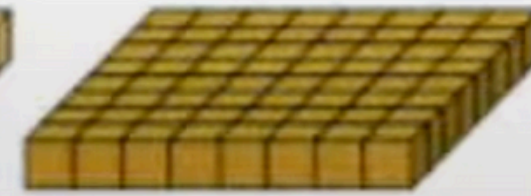
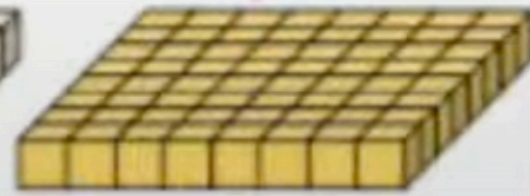
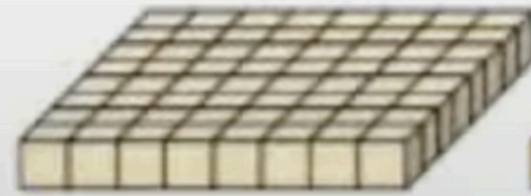
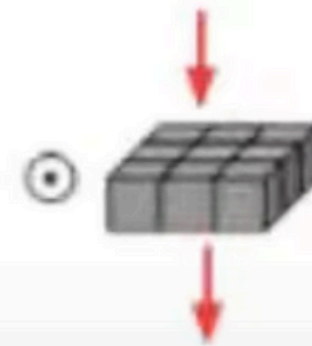
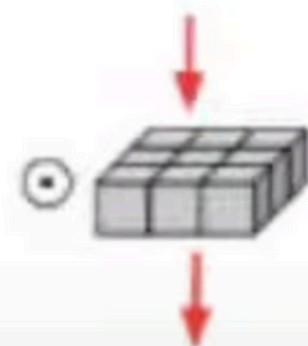
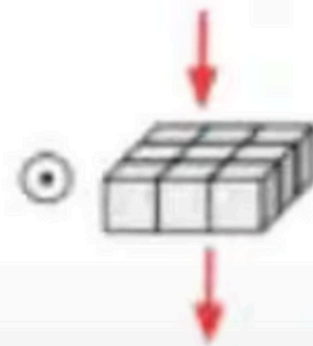
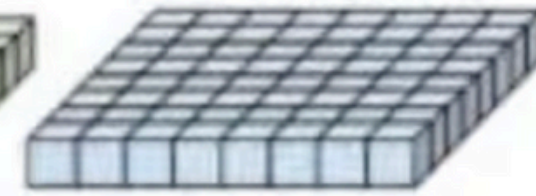
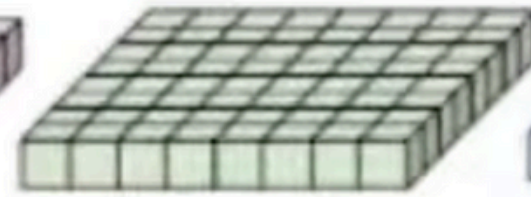
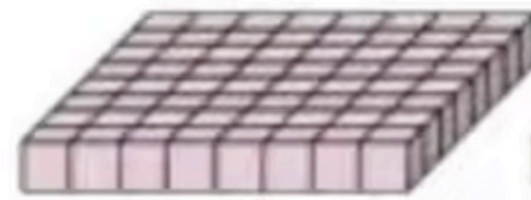
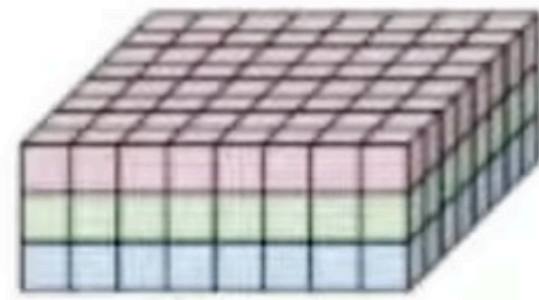
MOBILENET

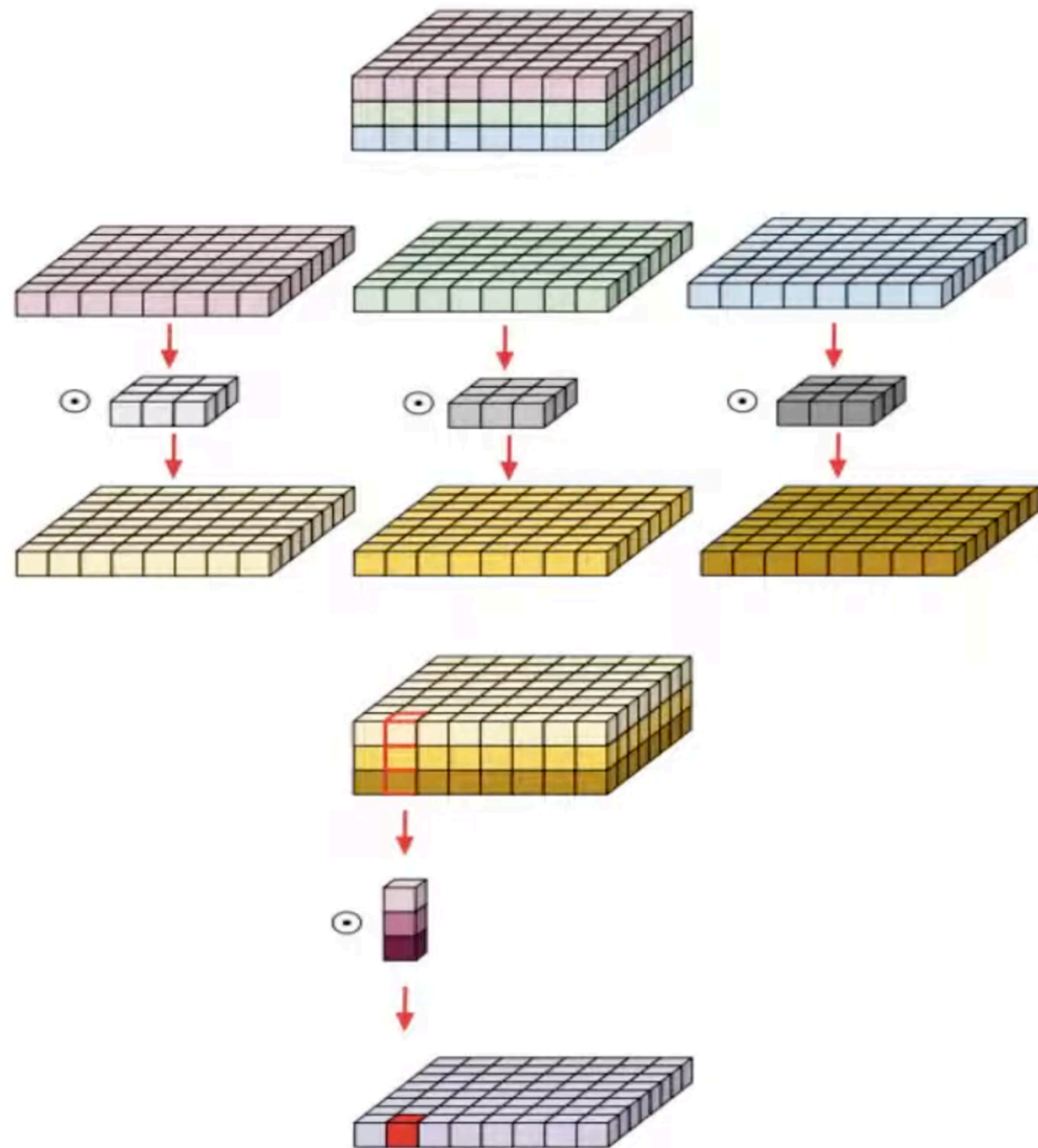
INTRODUCTION TO CNN



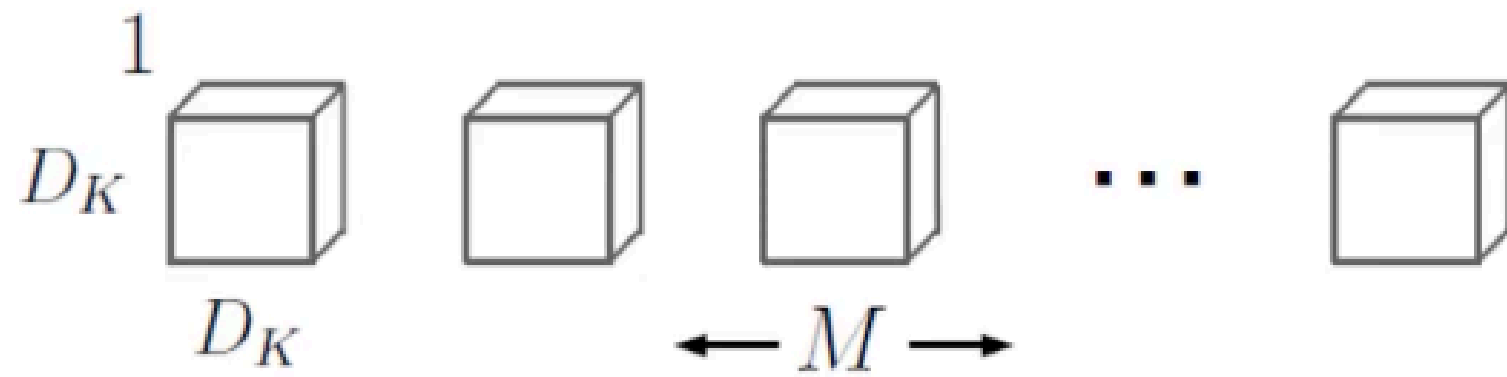


Df is 5x5 for example and M (channel) is 3. Then We have 2 filter which is 3x3 we need to multiply them all to see computational cost
5x5x3x2x3x3

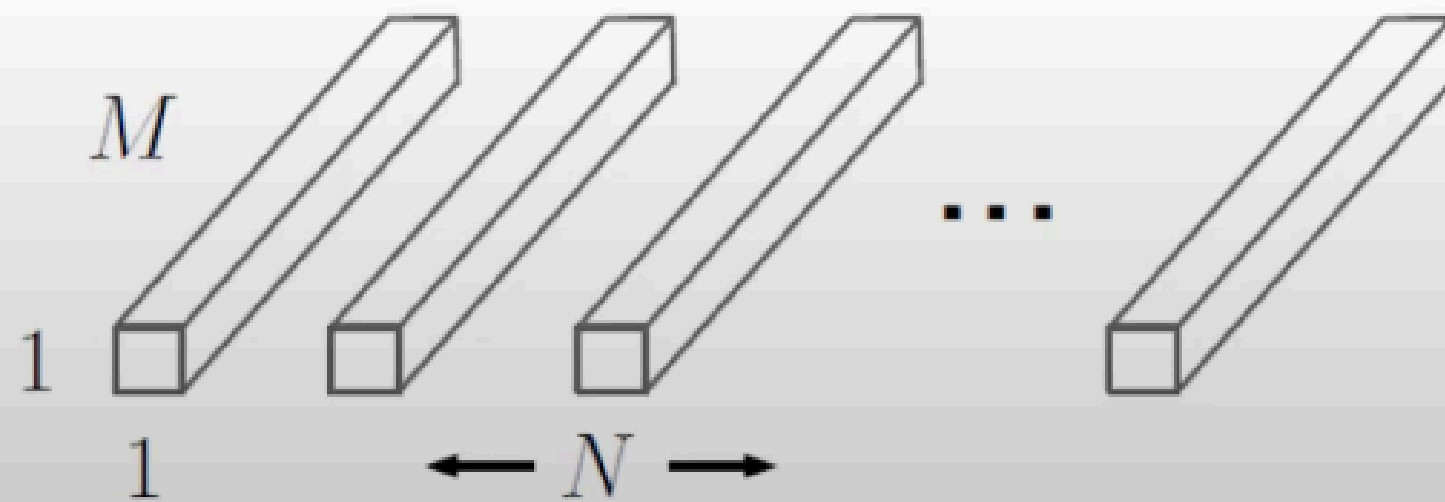




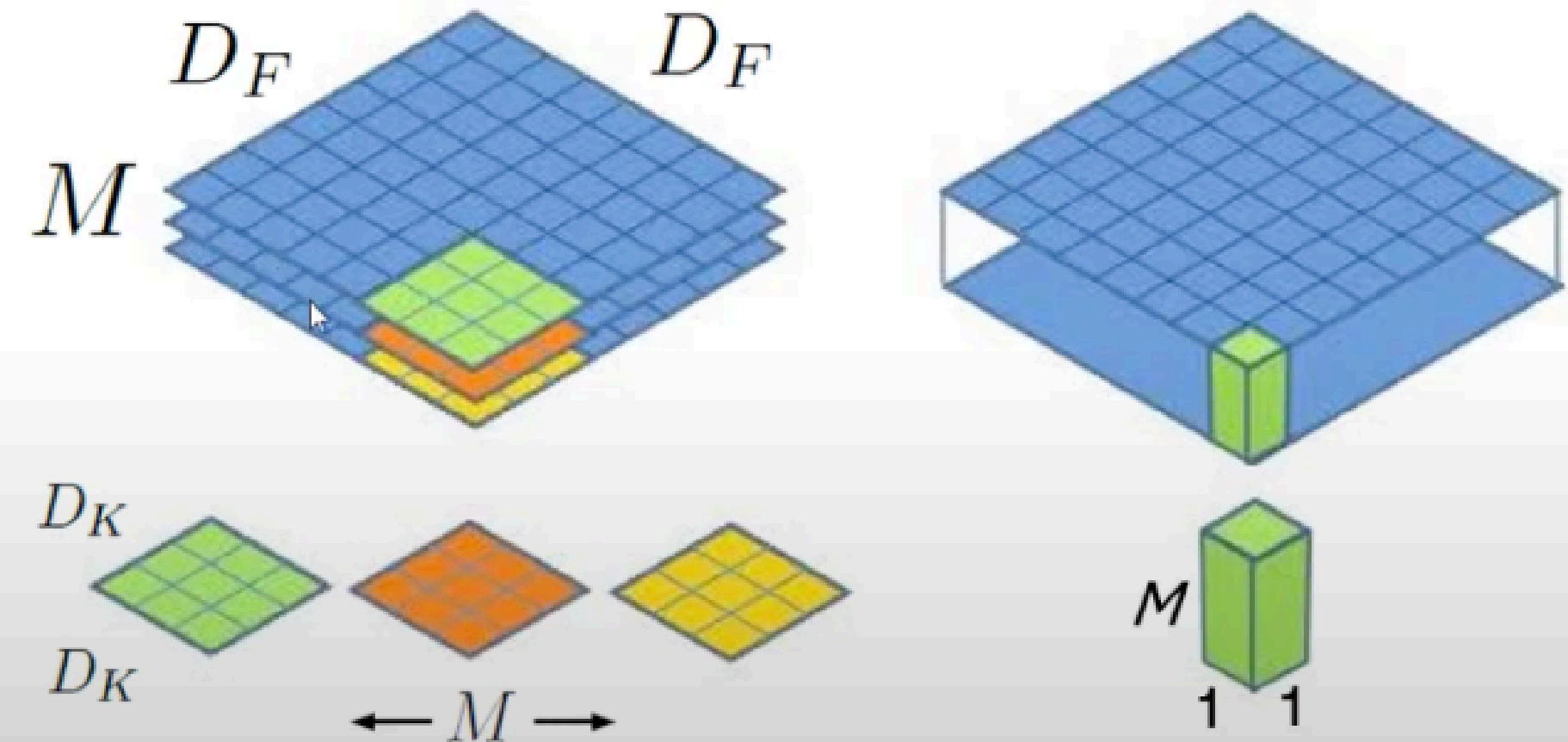
Depthwise Separable Convolution and Pointwise Convolution



(b) Depthwise Convolutional Filters



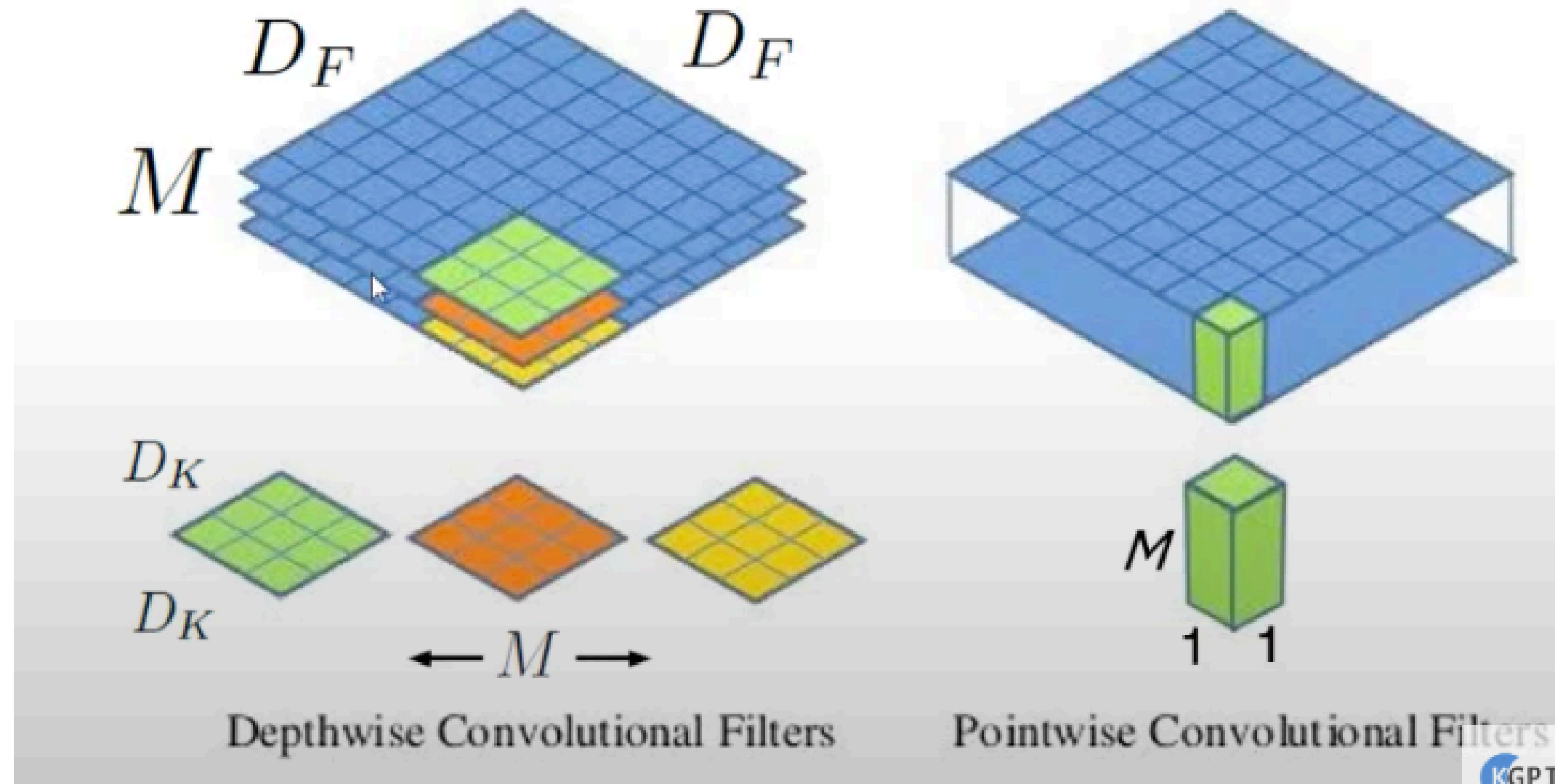
$$D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F$$



Depthwise Convolutional Filters

Pointwise Convolutional Filters

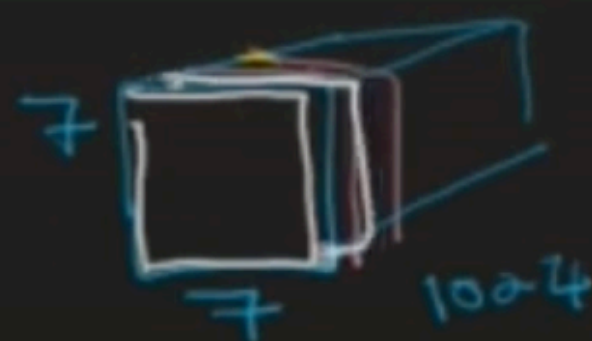
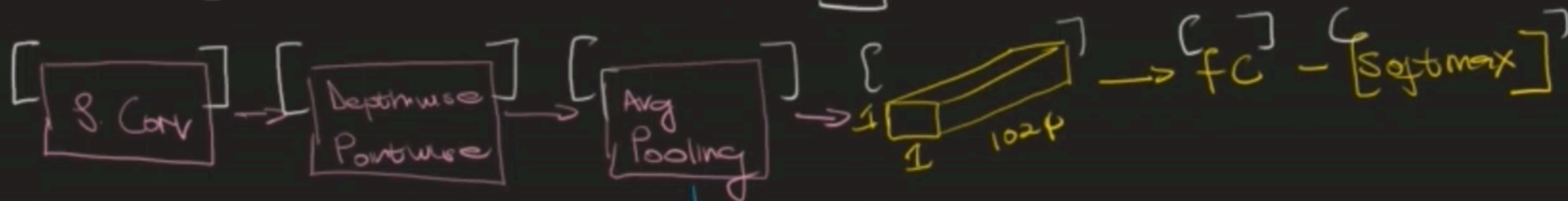
Normalde tüm filtreler tüm kanallara uygulanıyor 3 kanal varsa 5 filtremiz varsa toplamda 5 feature map oluşuyor.



MobileNet'de inputtaki her kanal için aynı bir filtre uygulanıyor. Diyelim RGB 3 kanal var, R'a bir filtre G'ye bir filtre B'ye bir filtre uygulayarak 3 feature map oluşturuyoruz.

Bu 1×1 filtreler 3 kanala uyguladığımız ayrı filtrelerden gelen 3 feature map'e bakarak yeni bir bilgi oluşturuyor. Kanallar arası etkileşimi sağlıyor.

Architecture of mobilenet V1



performed with a
window of $7 \times 7 \times 1$

of layer 28 layers

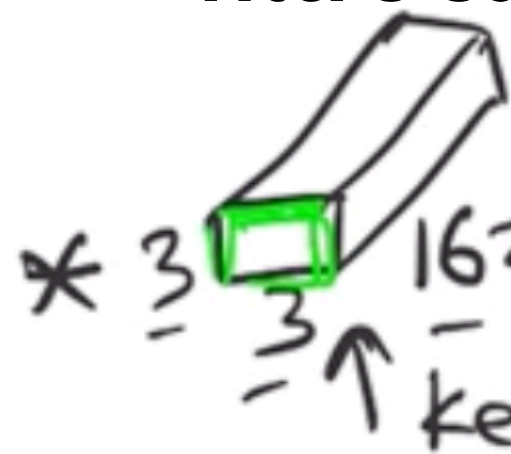
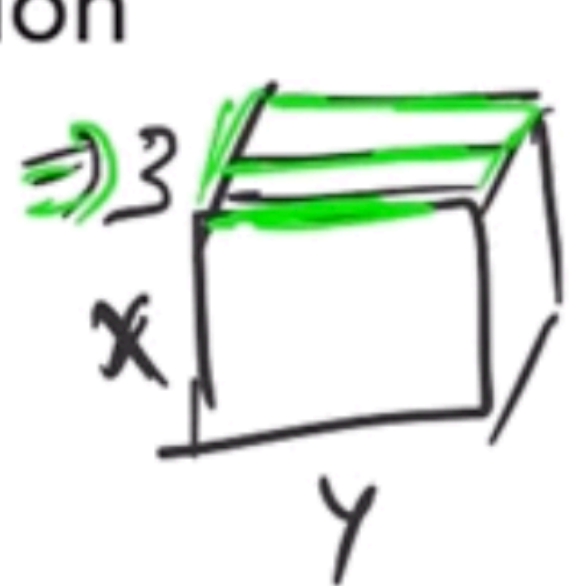
assuming DW and Pointwise are
separate layer

BN + RELU - all layer beside fc

ion

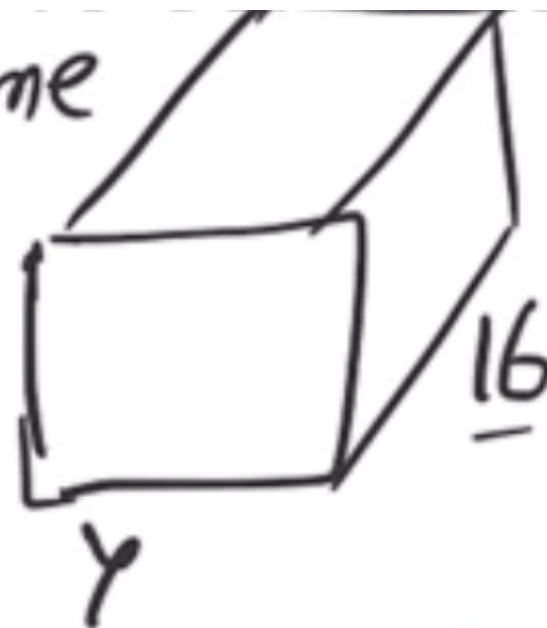
filtre sayısı

Padding = Same



#Channel
kernel size

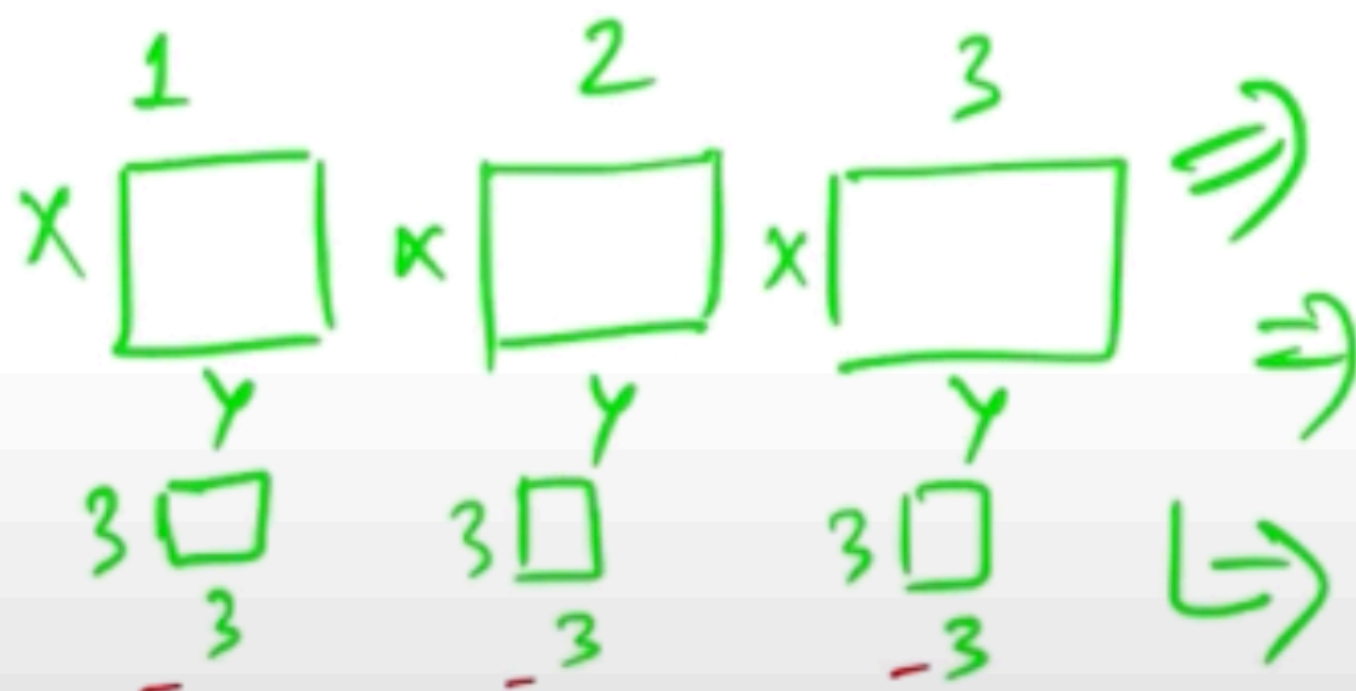
=>



#

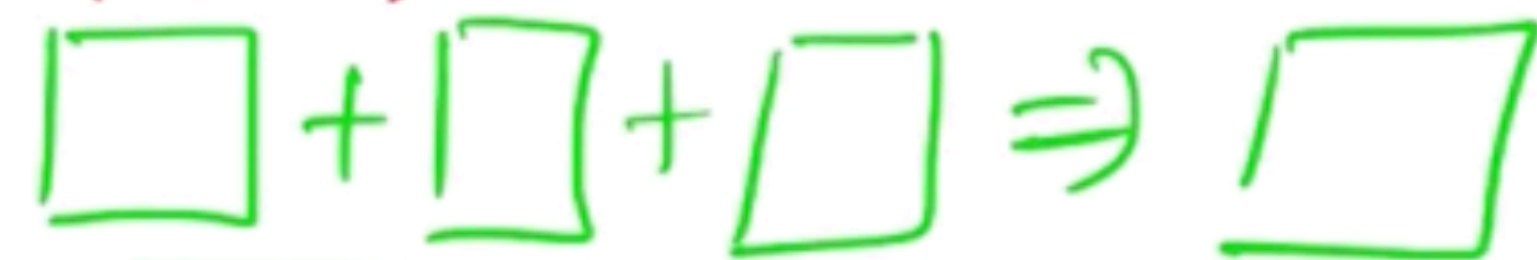
$$3 \times 3 \times 16 \times 3$$

$$= 432$$

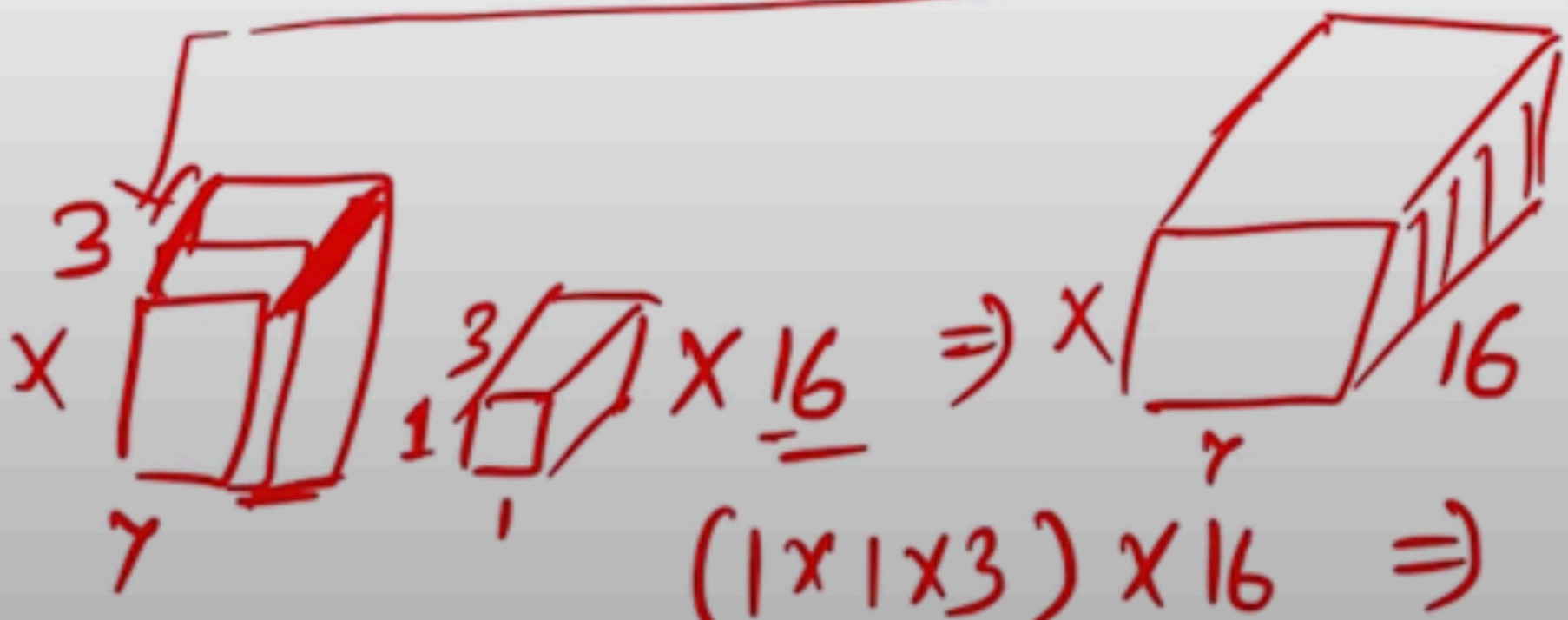


$$3 \times (3 \times 3) = 27$$

+ traditional

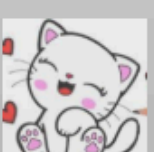


Concat Depth wise Conv



=> Depth + Pointwise Conv

$$27 + 48 = 75$$



$$\text{Computational Gain} = \frac{\text{Computational Cost of MobileNet}}{\text{Computational Cost of Standard Conv}}$$

$$\text{Computational Gain} = \frac{D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F}{D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F} = \frac{1}{N} + \frac{1}{D_K^2}$$

Number of filters and size of filters if they are both greater than 1 we always have computational gain.

Table 6. MobileNet Width Multiplier

Width Multiplier	ImageNet Accuracy	Million Mult-Adds	Million Parameters
1.0 MobileNet-224	70.6%	569	4.2
0.75 MobileNet-224	68.4%	325	2.6
0.5 MobileNet-224	63.7%	149	1.3
0.25 MobileNet-224	50.6%	41	0.5

Table 7. MobileNet Resolution

Resolution	ImageNet Accuracy	Million Mult-Adds	Million Parameters
1.0 MobileNet-224	70.6%	569	4.2
1.0 MobileNet-192	69.1%	418	4.2
1.0 MobileNet-160	67.2%	290	4.2
1.0 MobileNet-128	64.4%	186	4.2

Table 8. MobileNet Comparison to Popular Models

Model	ImageNet	Million	Million
	Accuracy	Mult-Adds	Parameters
1.0 MobileNet-224	70.6%	569	4.2
GoogleNet	69.8%	1550	6.8
VGG 16	71.5%	15300	138

Table 9. Smaller MobileNet Comparison to Popular Models

Model	ImageNet	Million	Million
	Accuracy	Mult-Adds	Parameters
0.50 MobileNet-160	60.2%	76	1.32
Squeezenet	57.5%	1700	1.25
AlexNet	57.2%	720	60

Table 10. MobileNet for Stanford Dogs

Model	Top-1	Million	Million
	Accuracy	Mult-Adds	Parameters
Inception V3 [18]	84%	5000	23.2
1.0 MobileNet-224	83.3%	569	3.3
0.75 MobileNet-224	81.9%	325	1.9
1.0 MobileNet-192	81.9%	418	3.3
0.75 MobileNet-192	80.5%	239	1.9

WHY DO WE NEED MOBILENET?

Smaller neural network size which means training fast

Easy to deploy on embedded devices

Low computational cost

Low latency

High accuracy